Phase 1 Report: Study Design and Descriptive Statistics

Association Between Sleep, Sedentary Behavior, and Systolic Blood Pressure (NHANES 2021–2023)

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1 Data Description and Study Design

1.1 Data Source and Data Collection

This project uses publicly available data from the National Health and Nutrition Examination Survey (NHANES), conducted by the National Center for Health Statistics (NCHS), a division of the U.S. Centers for Disease Control and Prevention (CDC). The selected data cycle covers August 2021 through August 2023.

NHANES is a nationally representative health survey designed to assess the health and nutritional status of the civilian, non-institutionalized population of the United States. It combines household interviews with physical examinations and other health measurements conducted in Mobile Examination Centers (MECs). Consequently, NHANES is an observational survey rather than a randomized experiment.

The focus of this project is on Investigating the Association between Lifestyle Factors (Sleep and Sedentary Activity) and Systolic Blood Pressure, and the Mediating Role of BMI.

The analysis will use the following public-use data files:

- `DEMO_L`: Demographic characteristics and survey design information.
- `SLQ_L`: Sleep-related questionnaire data.
- `PAQ_L`: Physical activity questionnaire data.
- `BPXO_L`: Oscillometric blood pressure measurements.
- `BMX_L`: Body measurements, including BMI.

The files are available through the official NHANES website: `https://wwwn.cdc.gov/nchs/nhanes/continuousnhanes/default.aspx?Cycle=2021-2023`.

The data were collected using multiple procedures. Demographic information and questionnaire responses were obtained through interviews, while blood pressure and body measurements were collected during physical examinations in Mobile Examination Centers. The physical activity questionnaire was administered by trained interviewers in the home or by telephone. Sleep duration was derived from participants' reported usual sleep and wake times.

1.2 Unit of Observation and Data Structure

The unit of observation in this project is an individual NHANES participant. The demographic, sleep, physical activity, blood pressure, and body measurement files contain participant-level records. The common identifier, `SEQN`, will be used to combine the selected files. Also datast contains at most one row per participant.

1.3 Sampling, Participant Selection, and Generalizability

NHANES uses a complex, multistage probability sampling design. The August 2021–August 2023 cycle used a stratified, clustered, four-stage sample design involving primary sampling units, area segments, dwelling units, and participants selected within households.

The sample was designed to represent the civilian, non-institutionalized population of the United States. However, participants did not all have identical probabilities of selection. Survey weights are therefore provided to account for unequal selection probabilities and to adjust for nonresponse and population coverage.

The present study is observational. No treatment or exposure was randomly assigned by the researchers. Therefore, associations between sleep duration, physical activity, BMI, and systolic blood pressure will not, by themselves, establish causal relationships.

1.4 Classification of the Main Variables

The primary outcome is systolic blood pressure, measured in millimeters of mercury (mmHg). The explanatory variables are usual sleep duration and physical activity. BMI will initially be considered a continuous variable and may also be used to define groups for descriptive comparisons. Age and sex will be included as demographic variables.

Table 1 summarizes the main variables and their measurement scales.

Variable	NHANES file	Measurement scale	Role and description
BPXOSY1(Systolic - 1st oscillometric reading)	BPXO_L	Quantitative, continuous	Primary response variable, measured in mmHg.three consecutive BP measurements (systolic and diastolic) were taken 60 seconds apart using a digital upper-arm electronic blood pressure measurement device.
SLD012(Sleep hours - weekdays or workdays)	SLQ_L	Quantitative, continuous	Explanatory variable. Number of hours usually sleep on weekdays or workdays.
PAD680(Minutes sedentary activity)	PAQ_L	Quantitative, continuous	Explanatory variable. How much time do you usually spend sitting on a typical day?
BMXBMI(Body Mass Index (kg/m**2))	BMX_L	Quantitative, continuous	explanatory or grouping variable. Adult BMI categories may be used for descriptive comparisons.
RIDAGEYR(Age in years at screening)	DEMO_L	Quantitative, discrete as recorded in years	Demographic variable.
RIAGENDR(Gender)	DEMO_L	Categorical, nominal	Demographic grouping variable.

Table 1: Preliminary classification of the main study variables.

1.5 Data cleaning and Missing values and Special tasks for Path 1

As you will see in the notebook, we merged the files using SEQN which is the unique identifier for each participant, Also inner join was used to keep only people who exist in all the files. Sample size before merge was 11933 people and afer merging it shrunk to 6337 people. After this we saw that missing values for each variable are as follows: BMI:102, Systolic Blood Pressure:214, Sleep Hours:67, sedentary activity:47. Note that these are before we delete participants with age under 20 and Also we treated codes like "Refused" and "Don't know" as missing values(they were encoded with numbers like 7777, 9999, 77, 99, etc).
Individuals with missing data for the primary response variable (Systolic Blood

Pressure), BMI, Sedentary activity, or Sleep Duration were dropped, Because the investigation without each of them is not complete and there are few so deleting them will not shrink the data too much. There is no repeated record. To delete records with age under 20 we drop 273 individuals. Then we delete records with missing key health data which are 350 individuals and so we are left with 5714 samples.

To analyze impossible or logical errors in we analyzed min and max variables for each variable as you will see in table 2 and they are all reasonable variables in the logical ranges.(e.g., Sedentary Min max is 1380 minutes = 23 hours, which is less than 1440 minutes/day, though quite high; Sleep Hours is between 2 and 14 hours; BMI and Systolic BP are within physiologically possible human ranges)

2 Descriptive Statistics

2.1 Frequency Tables and Measures of Center and Spread

For the categorical variable (Gender), the frequency table indicates a relatively balanced distribution between males and females in our analytical sample, Female counts is 3143 wiht 55.00 percent and Male counts 2571 with 45.00 percent. For the quantitative variables, the descriptive statistics are summarized in Table 2.
For roughly symmetric variables without extreme skew, Mean and Standard Deviation are appropriate, variables like Age and Sleep Hours, whereas for skewed variables or variables with notable outliers (like Sedentary Min where Mean is 363.93 vs. Median 300.0, or in BMI extreme high BMI values pull the Mean (29.74) above the Median, and to a lesser extent Systolic BP are right-skewed with upper outliers.), Median and IQR are more robust and appropriate measures of center and spread. Note that as mentioned in dataset documentation sleep hours less than 2 are recorded as 2 and sleep hours more than 14 are recorded as 14.

Table 2: Summary of Center and Spread for Quantitative Variables (Sample Data)

Variable	**Mean**	**Median**	**Std Dev**	**Min**	**Max**	**IQR**	**25%**	**75%**
Age (Years)	53.76	57.0	17.10	20	80	31.0	39.0	68
Systolic BP (mmHg)	123.02	121.0	18.48	75	225	24.0	110.0	133.0
BMI	29.74	28.4	7.16	11.1	74.8	9.2	24.7	33.5
Sleep (Hours)	7.70	8.0	1.57	2	14	2.0	7.0	8.5
Sedentary Activity(minutes)	363.93	300.0	208.81	0	1380	300.0	180.0	480.0

2.2 Primary Descriptive Camparison

Descriptively, male participants have a higher mean systolic BP (mean 125.61 mmHg, median 123.0 mmHg) compared to female participants (mean 120.86 mmHg, median 117.0 mmHg).

2.3 Visualization and Initial Descriptive Comparison

The figures below illustrate the distributions and initial relationships between the variables. The histogram of Systolic Blood Pressure (top-left) shows a roughly symmetric distribution with a slight right skew. The boxplot for Sleep Hours (top-right) highlights several outliers (e.g., sleep duration under 4 hours or over 11 hours), which will require attention in later modeling phases.

The scatter plot (bottom-left) displays a weak positive association between BMI and Systolic BP. In the descriptive comparison between groups (bottom-right), the median blood pressure for males appears slightly higher than for females.

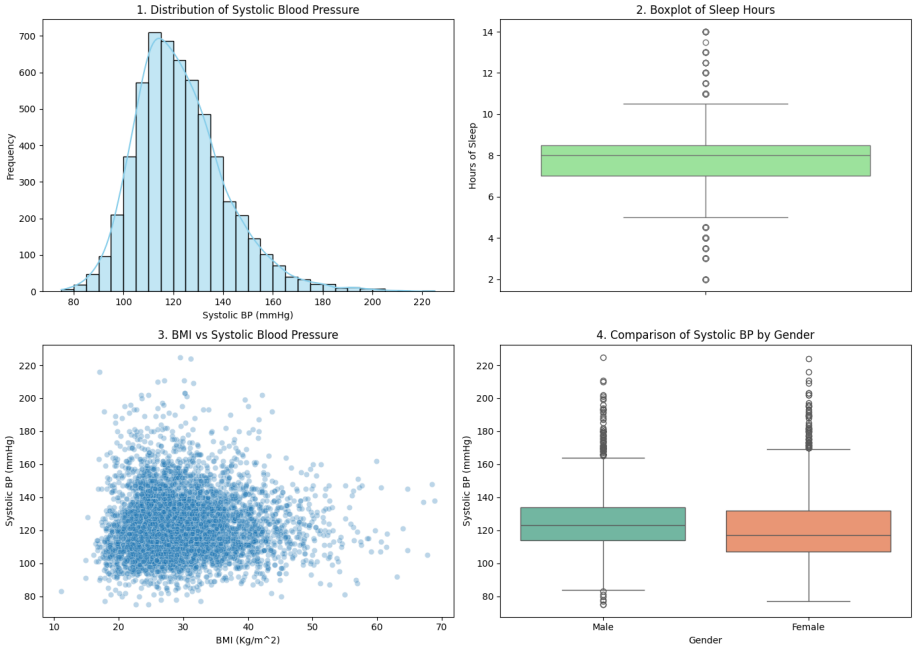


Figure 1: Visualizations of main variables and their relationships.

3 Sampling Design and Weights

Because NHANES oversamples certain demographic segments (such as older adults or specific racial minorities) to ensure reliable estimates, each individual

in the dataset represents a different number of people in the actual population. To account for this, the Medical Examination Center weight variable (`WTMEC2YR`) was identified.

To demonstrate the impact of the sampling design:

- **Raw (Unweighted) Mean** of Systolic BP was calculated directly from the sample : 123.00 mmHg
- **Weighted Mean** was calculated using `WTMEC2YR` : 121.22 mmHg
- **Raw (Unweighted) Mean** of Age was calculated directly from the sample : 53.75 years
- **Weighted Mean** was calculated using `WTMEC2YR` : 48.78 years

The observed difference between the raw mean and the weighted mean confirms that the "frequency in the sample" does not perfectly equal the "probability/estimate in the population." Using survey weights is crucial for accurately estimating true national population parameters.
Because there are too many older people in our raw sample of 5714 people, and older people have higher blood pressure, the simple average (Raw Mean) of our sample was be artificially too high. To fix this, CDC gives each person in the dataset a number called a sampling weight (WTMEC2YR).An oversampled older person might represent only 10,000 US adults(smaller weight). A younger adult might represent 60,000 US adults (larger weight).

4 Finalizing the Statistical Question

Based on the understanding gained from the data, the distributions of the variables, and the presence of high-quality clinical measurements, the statistical question is finalized as follows:

"Is there a significant association between daily sleep duration and daily sedentary behavior (minutes spent sitting) with the average systolic blood pressure in US adults? Furthermore, does this association hold when controlling for confounding variables such as age and BMI?"
With a final complete-case sample size of N=5714 adults and wide variation across sleep duration (2–14 hours), sedentary time (0–1380 minutes), and systolic BP (75–225 mmHg), the dataset provides sufficient statistical power and coverage to investigate this question.